DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW

Evaluation Scheme & Syllabus

for

M. Tech. Integrated 5th Year

Computer Science and Engineering

(Effective from the Session: 2023-24)

STUDY & EVALUATION SCHEME
M. Tech. Integrated
(Computer Science and Engineering) 5th YEAR

COURSE CURRICULUM
SEMESTER- IX

S. No.	Subject Codes	Subjects	Periods			Evaluation Schemes				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	MKCS951	Seminar-II	0	0	6	3			150		---	150	3
2	MKOE09X	Open Elective-III	3	0	0	30	20	50	---	100	---	150	3
3	MKCS952	Dissertation-I	0	0	28	---	---	---	150		300	450	14
		Total										**750**	**20**

OPEN ELECTIVE –III

MKOE093 ENTREPRENEURSHIP DEVELOPMENT
MKOE094 INTRODUCTION TO SMART GRID
MKOE095 QUALITY MANAGEMENT
MKOE096 INDUSTRIAL OPTIMIZATION TECHNIQUES

STUDY & EVALUATION SCHEME
M. Tech. Integrated
(Computer Science and Engineering) 5th YEAR

COURSE CURRICULUM
SEMESTER- X

S. No.	Subject Codes	Subjects	Periods			Evaluation Schemes				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	MKCS1051	Dissertation-II	0	0	36	---	---	----	300	---	600	900	18
		Total	**0**	**0**	**36**							**900**	**18**

MKOE093	ENTREPRENEURSHIP DEVELOPMENT	
	DETAILED SYLLABUS	**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	Entrepreneurship- definition. growth of small scale industries in developing countries and their positions vis-a-vis large industries; role of small scale industries in the national economy; characteristics and types of small scale industries; demand based and resources based ancillaries and sub-control types. Government policy for small scale industry; stages in starting a small scale industry	**08**
II	Project identification- assessment of viability, formulation, evaluation, financing, field-study and collection of information, preparation of project report, demand analysis, material balance and output methods, benefit cost analysis, discounted cash flow, internal rate of return and net present value methods.	**08**
III	Accountancy- Preparation of balance sheets and assessment of economic viability, decision making, expected costs, planning and production control, quality control, marketing, industrial relations, sales and purchases, advertisement, wages and incentive, inventory control, preparation of financial reports, accounts and stores studies.	**08**
IV	Project Planning and control: The financial functions, cost of capital approach in project planning and control. Economic evaluation, risk analysis, capital expenditures, policies and practices in public enterprises. profit planning and programming, planning cash flow, capital expenditure and operations. control of financial flows, control and communication.	**08**
V	Laws concerning entrepreneur viz, partnership laws, business ownership, sales and income taxes and workman compensation act. Role of various national and state agencies which render assistance to small scale industries.	**08**

Text Books:

1. Forbat, John, “Entrepreneurship” New Age International (P) Limited, 2019.
2. Havinal, Veerbhadrappa, “Management and Entrepreneurship” New Age Internationals, 2008.
3. Joseph, L.Masood, “ Essentials of Management”, 4th Edition, Prentice Hall of India, 2000.

Reference Books:

1. Arora M., Natarajan K. and Gordan E., “Entrepreneurship Development”, 1st ed; Himalaya Publishing House Pvt Ltd, 2009.
2. Janakiram B., Management & Entrepreneurship, Excel Books India, 2009.
3. Desai V., “The Dynamics of Entrepreneurial Development”, 5th ed; Himalaya Publishing House, 2013.

MKOE094	INTRODUCTION TO SMART GRID	
	DETAILED SYLLABUS	**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	Introduction: Introduction to Smart Grid: Evolution of Electric Grid, Concept of Smart Grid, Definitions, Need of Smart Grid, Functions of Smart Grid, Opportunities & Barriers of Smart Grid, Difference between conventional & smart grid, Concept of Resilient &Self-Healing Grid, Present development & International policies in Smart Grid. Case study of Smart Grid. CDM opportunities in Smart Grid.	**08**
II	Smart Grid Technologies: Introduction to Smart Meters, Real Time Prizing, Smart Appliances, Automatic Meter Reading (AMR), Outage Management System (OMS), Plug in Hybrid Electric Vehicles (PHEV), Vehicle to Grid, Smart Sensors, Home & Building Automation.	**08**
III	Smart Grid Technologies: Smart Substations, Substation Automation, Feeder Automation, Geographic Information System (GIS), Intelligent Electronic Devices (IED) & their application for monitoring & protection, Smart storage like Battery, SMES, Pumped Hydro, Compressed Air Energy Storage, Wide Area Measurement System (WAMS), Phase Measurement Unit (PMU), PMUs application to monitoring & control of power system.	**08**
IV	Microgrids and Distributed Energy Resources: Concept of microgrid, need & application of microgrid, formation of microgrid, Issues of interconnection, protection & control of microgrid, Plastic & Organic solar cells, thin film solar cells, Variable speed wind generators, fuel cells, microturbines, Captive power plants, Integration of renewable energy sources.	**08**
V	Power Quality Management in Smart Grid: Power Quality & EMC in Smart Grid, Power Quality issues of Grid connected Renewable Energy Sources, Power Quality Conditioners for Smart Grid, Web based Power Quality monitoring	**08**

Text Books:

1. Ali Keyhani, Mohammad N. Marwali, Min Dai, “Integration of Green and Renewable Energy in Electric Power Systems”, Wiley, 2009.
2. Clark W. Gellings, “The Smart Grid: Enabling Energy Efficiency and Demand Response”, CRC Press, 2009.
3. Janaka Ekanayake, Nick Jenkins, KithsiriLiyanage, Jianzhong Wu, Akihiko Yokoyama, “Smart Grid: Technology and Applications”, Wiley, 2012.
4. Nouredine Hadjsaid, Jean-Claude Sabonnadiere (Editor) “Smart Grids”, John Wiley & Sons Ltd, May 2012.
5. Stuart Borlase, “Smart Grids: Infrastructure, Technology, and Solutions”, CRC Press, Taylor & Francis Group, 2013.

Reference Books:

1. Andres Carvallo, John Cooper, “The Advanced Smart Grid: Edge Power Driving Sustainability”,Artech House Publishers, July 2011.
2. James Northcote, Green, Robert G. Wilson, “Control and Automation of Electric Power Distribution Systems (Power Engineering)”, CRC Press, 2007.
3. MladenKezunovic, Mark G. Adamiak, Alexander P. Apostolov, Jeffrey George Gilbert, “SubstationAutomation (Power Electronice and Power Systems)”, Springer, 2014.
4. R.C. Dugan, Mark F. McGranghan, Surya Santoso, H. Wayne Beaty, “Electrical Power System Quality”, 2nd Edition, McGraw Hill Publication, 2012.

MKOE095	QUALITY MANAGEMENT	
	DETAILED SYLLABUS	**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	Quality Concepts: Evolution of Quality Control, concept change, TQM Modern concept, Quality concept in design, Review of design, Evolution of proto type. Control on Purchased Product: Procurement of various products, evaluation of supplies, capacity verification, Development of sources, procurement procedure. Manufacturing Quality: Methods and techniques for manufacture, inspection and control of product, quality in sales and services, guarantee, analysis of claims.	**08**
II	Quality Management: Organization structure and design, quality function, decentralization, designing and fitting, organization for different type products and company, economics of quality value and contribution, quality cost, optimizing quality cost, seduction program. Human Factor in quality Attitude of top management, cooperation of groups, operators attitude, responsibility, causes of apparatus error and corrective methods.	**08**
III	Control Charts, Theory of control charts, measurement range, construction and analysis of R charts, process capability study, use of control charts. Attributes of Control Chart, Defects, construction and analysis of charts, improvement by control chart, variable sample size, construction and analysis of C charts	**08**
IV	Defects diagnosis and prevention defect study, identification and analysis of defects, correcting measure, factors affecting reliability, MTTF, calculation of reliability, building reliability in the product, evaluation of reliability, interpretation of test results, reliability control, maintainability, zero defects, quality circle.	**08**
V	ISO-9000 and its concept of Quality Management, ISO 9000 series, Taguchi method, JIT in some details.	**08**

Text Books:

1. Lt. Gen. H. Lal, “Total Quality Management A Practical Approach”, Eastern Limited, 1990.
2. Gregory Bounds, “Beyond Total Quality Management: Toward the Emerging Paradigm”, McGraw Hill, 1994.
3. Menon, H.G, “TQM in New Product Manufacturing”, McGraw Hill, 1992.

Reference Books:

1. James R Evans and William M Lindsay, "The Management and Control of Quality", South Western College Pub., 2010.
2. Oakland J S, "Total Quality Management – Text with Cases", A Butterworth-Heinemann Title, 2003.

MKOE096	INDUSTRIAL OPTIMIZATION TECHNIQUES	
	DETAILED SYLLABUS	**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**I Linear Programming:** Historical development of optimization, engineering application of optimization, formulation of design problems as a mathematical programing problem. Graphical method of solution, Simplex method, Dual Simplex method and its application in engineering. **Transportation and Assignment**: Introduction, Mathematical formulations, optimal solution of transportation model. Assignment problems: mathematical formulation, solution of Assignment models (Hungarian method), variation of the Assignment problem, the travelling salesman problem and their application in Engineering	**08**
II	**Sequencing and Network Analysis:** Introduction of sequencing, General assumptions, n Jobs through 2 machines, n jobs through 3 machines, n jobs through m machines, 2 jobs through m machines and their applications in Engineering. **Network Analysis: Introduction,** Network logic (Network or arrow diagram), Rules for drawing network diagrams, time analysis, forward and backward computation CPM and PERT, and their applications in Engineering.	**08**
III	**Theory of Games and Queueing Models:** Introduction, 2 person zero sum games, Maximin and minimax principle, game with saddle point and without saddle point, Principle of dominance, Rectangular games, graphical solution of 2xn or mx2 games. **Queuing model:** Introduction, Application of Queuing model, generalized Poisson queuing model, single server models and multiple channel Queuing modeland their applications in Engineering.	**08**
IV	**Dynamic Programming and Simulation:** Introduction Formulation of Dynamic Programming Problem, Dynamic Programming Algorithm, Forward recursions, Capital Budgeting Problem, Cargo-loading Problem. Solution of LPP by DPP **Simulation:** Introduction, definition and types of simulation, need for Simulation advantage and disadvantage, application of simulation, simulation procedure, Monte Carlo simulation and their applications in Engineering	**08**
V	**Inventory Control and Replacement Models:** Introduction, types of inventories, Inventory cost, Deterministic and probabilistic (nondeterministic) inventory models and their application in engineering. **Replacement models:** Introduction, definition, Replacement of items that deteriorate, Replacement of items that fail suddenly, Equipment Renewal Problem, Individual and Group Replacement policies & their applications in Engineering	**08**

Text Books:

1. Singiresu S. Rao., “Engineering Optimization: Theory and Practice”. New Age International, New Delhi, 2013.
2. R. Panneerselvam., “Operations Research”, Prentice- Hall of India, New Delhi, 2016.
3. Eliezer Naddor., “Inventory Systems”. John Wiley & Sons, Inc. New York, 1966.

Reference Books:

1. H.A. Taha, “Operations Research: An Introduction”, Macmillan Publishing Company, Inc., New York, 2019.
2. K. Swarup, P.K. Gupta, M. Mohan, “Operations Research”, Sultan Chand and Sons, New Delhi, 2014.
3. P.K. Gupta, D.S. Hira, “Operations Research” – An Introduction, S. Chand & Company Limited, New Delhi, 2015.
4. S.S. Rao, “Optimization Theory and Applications”, Wiley Eastern Ltd., New Delhi, 1980.
5. J.K. Sharma, “Operations Research: Theory and Applications”, Mac Millan India, 2012.

M.Tech Integrated(CSE) 5th YEAR	
Subject Code- MKCS951	**L - T - P** 0-0-6
Subject Name-Seminar-II	**Credits** **3**

- Each M.Tech Int. student registered for seminar-II will give a 30-40-minute seminar in the form of a presentation based on his/her research work explaining his/her work.
- In addition, they will be expected to attend all seminars taken by the industry/ academic experts in their specialization scheduled in that semester on emerging trends and technology.
- Grading for seminar shall be dependent on the presentation of the work done and based on the quality of the seminar presented by the student and his/her attendance in other seminars.
- The faculty coordinator will be the nominal instructor for the course.
- However, grades will be given in consultation with thesis supervisor(s) of the concerned student.

M.Tech Integrated(CSE) 5th YEAR	
Subject Code- MKCS952	**L - T - P** 0-0-28
Subject Name- Dissertation-I	**Credits** **1**4

- The title of thesis should be the same as approved by the supervisor, and Performance Evaluation Committee (PEC) of the institute.
- The research scholar will earn 14 credits for the course.
- The research scholar should publish at least two research papers related to the approved topic of research in national/international journals of repute (SCI/SCOPUS/SCOPUS Conference).
- The research scholar should complete the research work and prepare a draft thesis which should be duly checked and certified by the approved supervisor(s) and PEC.
- Submit draft thesis or report at the end of semester containing summary of research work done and future work in dissertation-II in soft binding at least two weeks in advance.
- Prepare a presentation on the draft thesis of approximately 30-40 minutes using Power Point slides. The presentation must cover the following:
- Objectives and scope of the study
- Literature review (briefly 3-5 slides only)
- Identification of the research gap (s) based on literature review
- Problem formulation
- Research methodology
- Experimentation/Survey
- Data collection, analysis, and interpretation
- General conclusions
- Specific conclusions/contributions of the research
- Contribution of the research to society
- Further scope of research
- If you'll need any special arrangement/equipment for the presentation, inform the college/department/centre well in advance.
- Note the corrections/suggestions made during and after the pre thesis submission presentation for necessary action.
- Collect the necessary proforma and the procedure for thesis submission.

M.Tech Integrated(CSE) 5th YEAR	
Subject Code- MKCS1051	**L - T - P** 0-0-36
Subject Name- Dissertation-II	**Credits** **18**

- The title of thesis should be exactlythe same as approved for Dissertation-I.
- The research scholar must have earned at least 14 credits in theDissertation-I.
- The research scholar must have published at least two research papers related to the approved topic of research in national/international journals of repute (SCI/SCOPUS/SCOPUS Conference).
- The research scholar should clear all the outstanding dues including fine, if any.
- The research scholar should complete the research work and prepare a final thesis which should be duly checked and certified by the approved supervisor(s). All the Chapters should be ready and properly arranged, Figures, Tables are properly numbered, and standard referencing guidelines as per APA format should be followed. Kindly make sure that there are no typographical, referencing and formatting errors.
- Submit five hard binding copies of thesis and one soft binding containing the thesis, ppt, and research paper at least a month in advance.
- Request a mutually convenient date forthesis submission presentation in consultation with Director/Principal/Head of the concerned College/Department/Centre
- Prepare a presentation on the thesis of approximately 45 to 60 minutes using Power Point slides. The presentation must cover the following:
- Objectives and scope of the study
- Literature review (briefly 3-5 slides only)
- Identification of the research gap (s) based on literature review
- Problem formulation
- Research methodology
- Experimentation/Survey
- Data collection, analysis, and interpretation
- General conclusions
- Specific conclusions/contributions of the research
- Contribution of the research to society
- Further scope of research
- If you'll need any special arrangement/equipment for the presentation, inform the college/department/centre well in advance.
- Note the corrections/suggestions made during and after the pre thesis submission presentation for necessary action.
- Collect the necessary proforma and the procedure for thesis submission.